Chapter 2: Variables, Expressions, and Statements

1. Constants (Literals)

Constants are fixed values that do not change during program execution.

- **Numeric**: Integers (123) or Floats (98.6).
- **String**: Text wrapped in single (') or double (") quotes.
- **Boolean:** can be (0/False) or (1/True)

2. Variables & Memory

A variable is a **named location** in memory where you store data.

- **The Rules**: Must start with a letter or underscore _; can only contain letters, numbers, and underscores; and is strictly **case-sensitive**.
- **Mnemonic Names**: Best practice is to use "memory aid" names (e.g., hours instead of x) so humans can understand the code's intent.
- **Assignment**: The = sign is an **assignment statement**, not a mathematical "equals." Python evaluates the right side first, then stores the result in the variable on the left.

```
age = 15
age += 1          # Increment x by 1
age = age - 2       # decrement x by 2
print(age)        # Outputs: 14
```

3. Data Types: The type() Function

Python tracks the "class" of every value.

value	type
"Hello World"	String
123, 234	Integer
23.67	Float
True / False	Boolean

4. Operators & Precedence (PEMDAS)

Python follows a strict "Order of Evaluation" to decide which operation happens first.

Operator	Operation	Example
+	Addition	x + 2
-	Subtraction	x - 2
*	Multiplication	x * 2
/	Division	x / 2 (always results in a float)
//	Floor Div	10 // 3 (always results int)
**	Power (Exponentiation)	x ** 3
%	Remainder (Modulus)	x % 2 (useful for finding even/odd)

Note:

- Operations evaluate left to right, following precedence: () → ** → * / → + -
- **Cannot** add *strings* to *integers* directly without conversion.

5. Type Conversion & Input

- **input()**: Pauses the program to get user data. **Crucial**: It always returns a **string**, even if the user types a number.
- **Explicit Conversion**: Use int(), float() or bool() to convert string input into a numeric type before doing math, can also use str().
- **Implicit Conversion**: If you mix an int and float in a calculation, Python converts the result to a float automatically.

6. Documentation & Debugging

- **Comments**: Use # to write notes for humans. Python ignores everything after the # on that line .
- **Reserved Words**: You cannot name a variable any of the 35 keywords (like if, while, lambda) .
- **Common Errors**:
    - **NameError**: Using a variable before it has been assigned or misspelling it.
    - **TypeError**: Trying to "add" a string to an integer without conversion.